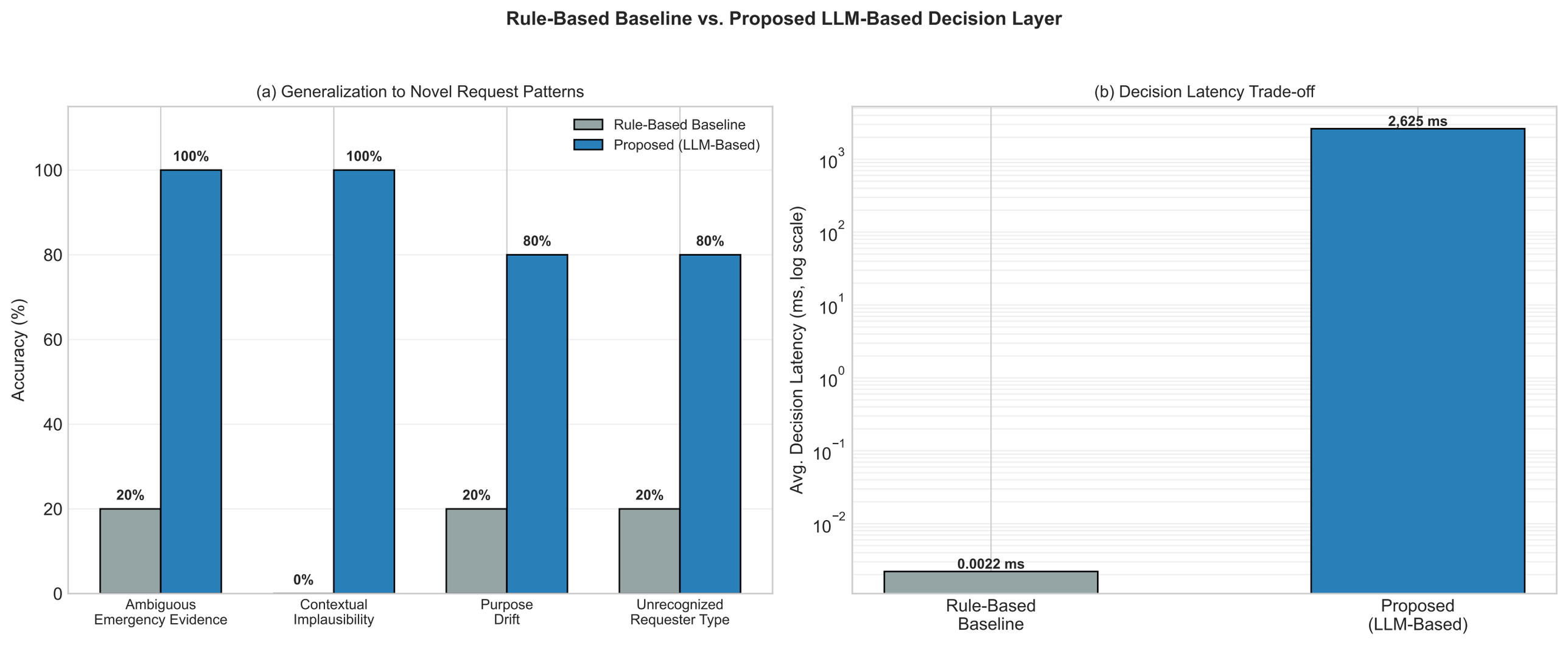
Rule-Based Baseline vs. Proposed LLM-Based Decision Layer
(a) Generalization to Novel Request Patterns
Rule-Based Baseline
Proposed (LLM-Based)
Accuracy (%)
0
20
40
60
80
100
20%
100%
0%
100%
20%
80%
20%
80%
Ambiguous Emergency Evidence
Contextual Implausibility
Purpose Drift
Unrecognized Requester Type
(b) Decision Latency Trade-off
Avg. Decision Latency (ms, log scale)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
0.0022 ms
2,625 ms
Rule-Based Baseline
Proposed (LLM-Based)